

Generation Interconnection System Impact Study Report

For Queue Project Q12-0147

Monfort 100 kV
February 15, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0147. The Interconnection Customer proposes a 58.1 MW wind facility interconnecting at the Monfort 100 kV substation in the Indiana area. The request was received on July 31, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 106 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,440,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Blackrock - Shawville Station 345 KV, the Carbon Ctr 500/345kV TX loads to 109.8% of its 523 MVA rating. Project DFAX: 7.3%.

The Waynedale Substation - Aep Pettit Avenue 100 KV Line 1 is overloaded to 104.2% of its 152 MVA emergency rating for the loss of the Waynedale - State St 230 KV. The project contributes 7.8% DFAX (4.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carbon Ctr 500/345kV TX	Replace transformer with higher-rated unit	\$33,424,000	1% (\$427,000)
Waynedale Substation - Aep Pettit Avenue 100 KV Line 1	Reconductor line	\$1,013,000	100% (\$1,013,000)
	Total allocated to project		\$1,440,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.